

PREIMAGE-CHAIN CORRECTIONS, PRIMITIVE-BLOCK RIGIDITY, AND NEW LOWER BOUNDS FOR SUPERPERMUTATIONS

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ABSTRACT. Building on the Lean-verified component method of Hunter and Raudvere, this paper quantifies the additional cost imposed when a spanning system of strongly exitless components is realized as the Hunter image of a Hamilton path in the permutation-overlap graph.

A component-matching theorem for the auxiliary preimage-chain graph identifies its source and target components up to one global exception. At the source end, all Hunter components are represented except for the source/terminal exception. At the target end, all non-root Hunter components admit a capacity-one matching, augmented by a terminal symbol when required. This structure yields an exact *preimage-chain identity* that decomposes $\text{wt}(P) - B_1(F(P))$ into nonnegative root, terminal, chain, and cycle terms. A shortest-path and assignment relaxation defines a computable functional Π satisfying

$$\text{wt}(P) \geq B_1(F(P)) + \Pi(F(P)).$$

For $k = 6$, an explicit Hunter image has $B_1 = 863$ and $\Pi = 6$, whereas a known 872-letter construction has $B_1 = 866$ and $\Pi = 0$.

A σ^2 -reduced normal form, a rate-defect normal form, and an endpoint-matching capacity theorem are obtained for exact weight-three interval-piece chains. They imply the primitive-block stability inequality

$$(k-2)t \leq (k-2)D_k\pi + (D_k-1)d, \quad D_k = k^2 - 3k + 1,$$

where $\pi = 1$ precisely when a component has minimum entry weight two and terminates in a full interval piece. In a reduced preimage, the associated terminal portal cannot be absorbed at zero cost by any minimum-entry-two target, including a self-nested target.

The pointwise actual-image Hunter bound assigns a canonical nonnegative integer layer to every Hamilton path. Portal rigidity, defect-budget monotonicity, the root-biased distinguished-component envelope, and capacity-one target matching then give

$$S(k) \geq \mathcal{H}_k + \Gamma_k, \quad \Gamma_k = \left(1 - \frac{3}{k} + O(k^{-2})\right) (k-4)!.$$

In particular,

$$S(7) \geq 5892, \quad S(8) \geq 46118, \quad S(14) \geq 93,890,256,441.$$

The arbitrary- k proof chain, including actual component capacity, the pointwise actual-image baseline, the portal assignment, σ^2 normalization, the unconditional pathwise theorem, the global $S(k)$ theorem, and the numerical corollaries for $k = 5, \dots, 14$, has been formalized in Lean 4.31.0 against a pinned Hunter–Raudvere library. The formal development contains no unfinished proof or project-local axiom. A separate deterministic Python implementation reproduces the finite certificates, local transcriptions, and exact integer tables.

An explicit two-block bridge family further shows that every single-component, single-portal linear envelope of the form $t(C) \leq D_k\pi(C) + c_k d(C)$ must satisfy $c_k = \Omega(k)$ for $k \geq 8$. This obstruction does not apply to nonlinear, multistate, prefix-sensitive, intercomponent, or iterated-transform methods.

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1. INTRODUCTION

A *superpermutation* on k symbols is a finite word over a k -letter alphabet in which every permutation of the alphabet occurs as a contiguous factor. Let $S(k)$ denote the minimum possible length. The values $S(1), \dots, S(5)$ are known, whereas the problem remains open for every $k \geq 6$. Ashlock and Tillotson gave the classical recursive construction of length $\sum_{i=1}^k i!$ [AT93]; Johnston established non-uniqueness at the formerly conjectured length [Joh13]; and Houston constructed a six-symbol superpermutation of length 872 [Hou14]. Egan’s construction, based on Williams’s Hamilton path in a two-generator Cayley digraph [Wil17, Ega18], yields

$$(1) \quad S(k) \leq k! + (k-1)! + (k-2)! + (k-3)! + k - 3.$$

A formally verified gain-one lift lowers this upper bound by one for every $k \geq 8$ [Rau26].

The present work builds on the lower-bound framework of Hunter and Raudvere [HR26], whose complete theorem chain has been formalized in Lean. Define

$$(2) \quad D_k = k^2 - 3k + 1, \quad A_k = k^3 - 2k^2 - k - 1.$$

Their exact pointwise lower bound for superpermutation length is

$$(3) \quad \mathcal{H}_k = k! + (k-1)! + (k-2)! + k - 3 + \left\lceil \frac{(k-2)! - (k-2)}{D_k} \right\rceil.$$

In particular, $\mathcal{H}_6 = 869$, $\mathcal{H}_7 = 5888$, and $\mathcal{H}_8 = 46103$. The same formal development characterizes the image of the Hunter transformation as the class of spanning unions of strongly exitless paths; this characterization is essential in the six-symbol analysis below.

Hunter's bound prices the transformed components separately. The central question is therefore not whether the estimate for an isolated strongly exitless component can be sharpened, but how much additional cost is imposed by the simultaneous realization of several such components as the image of a single Hamilton path. The analysis consists of the following components.

- (1) A preimage-chain component-matching theorem, an exact identity for $\text{wt}(P) - B_1(F(P))$, and a computable shortest-path/assignment relaxation II.
- (2) A global σ^2 -reduced normal form that replaces every zero-cost decomposable double rotation by a proper door without increasing path weight.
- (3) The rate-defect normal form

$$d = D_k(x + \mu - 2) + (k-2)\Delta - r, \quad t = D_k m - (k-1)d.$$

- (4) An endpoint-matching capacity theorem and the primitive-block stability inequality, including the self-nested terminal-portal case.
- (5) A pointwise actual-image Hunter baseline that assigns every Hamilton path a canonical nonnegative integer layer $s(P)$.
- (6) An explicit layerwise correction derived from terminal-portal rigidity, with one global exception and any distinguished non-root target charged to the exact root gap.
- (7) Equality of the actual Hunter image class and a locally defined relaxed component class, established through the formalized image characterization.
- (8) Two limitations of the method: for $k \geq 8$, a two-block bridge family forces $c_k = \Omega(k)$ in every envelope $t(C) \leq D_k \pi(C) + c_k d(C)$, while neutral weight-three refinement creates no independent block-level obligation.

The main numerical consequences are summarized below.

Theorem 1.1 (Lean-formalized numerical corollaries). *The unconditional formal theorem gives*

$$\begin{array}{ll} S(5) \geq 153, & S(6) \geq 869, \\ S(7) \geq 5,892, & S(8) \geq 46,118, \\ S(9) \geq 408,418, & S(10) \geq 4,033,080, \\ S(11) \geq 43,916,235, & S(12) \geq 522,610,764, \\ S(13) \geq 6,746,523,219, & S(14) \geq 93,890,256,441. \end{array}$$

The values for $k = 7, \dots, 14$ are the new lower bounds of this paper; $k = 5, 6$ are included to state the complete formally exported table.

The arbitrary- k structural results in the main proof chain are formalized in Lean 4.31.0. The formal development is published in the

[public GitHub repository](#).

The exact formalization source snapshot is commit

a3852f0de2d84b99cc65d4c281e86892b5f6e911.

A separate deterministic Python implementation reproduces the finite transcriptions and certificate checks; its role is computational audit rather than a second theorem-prover formalization or external peer review.

2. THE OVERLAP GRAPH AND HUNTER'S COMPONENT CLASSES

Let $\mathfrak{S}_k$ be the symmetric group written as words of length k . For $u, v \in \mathfrak{S}_k$, define

$$(4) \quad w(u, v) = \min\{d \in \{1, \dots, k\} : u_{d+1} \cdots u_k = v_1 \cdots v_{k-d}\}.$$

For a Hamilton path $P = (p_1, \dots, p_k)$, put

$$\text{wt}(P) = \sum_{i=1}^{k!-1} w(p_i, p_{i+1}).$$

The corresponding word has length $k + \text{wt}(P)$, and hence

$$(5) \quad S(k) = L(k) + k,$$

where $L(k)$ is the minimum Hamilton-path weight.

Write a word as $u = abR$, with $|R| = k - 2$. Define

$$(6) \quad \sigma(u) = bRa, \quad \sigma^2(u) = Rab, \quad \tau(u) = Rba.$$

Thus σ is the weight-one rotation and τ is the proper weight-two door. The σ -orbits are the *rotation classes*; each has k vertices, and there are $(k-1)!$ classes.

An *early exit* occurs when a path leaves a rotation class along an edge of weight at least two before visiting the weight-one successor of the current vertex. A path with no early exit is *exitless*; it is *strongly exitless* if, in addition, the weight-one successor of its final vertex already lies on the path.

Let $\mathcal{X}_k$ denote the actual image class

$$\mathcal{X}_k = \{F(P) : P \text{ is a Hamilton path}\}.$$

Let $\mathcal{Y}_k$ denote the class of spanning directed graphs whose weak components are vertex-simple strongly exitless paths. The following statement combines the necessity and sufficiency directions formalized by Hunter and Raudvere.

Theorem 2.1 (Hunter–Raudvere image characterization and component bound). *For $k \geq 2$,*

$$\boxed{\mathcal{X}_k = \mathcal{Y}_k.}$$

For every Hamilton path P and $X = F(P)$:

- (i) *every component of X is a union of complete rotation classes;*
- (ii) *the tail t_C of a component C is the first vertex of C visited by P ;*
- (iii) *with $\mu_C = \text{minto}(C)$,*

$$(7) \quad \text{wt}(P) \geq B_1(X) := |\text{Comp}(X)| - 1 + \text{wt}(X) + \sum_C \mu_C - \max_C \mu_C;$$

- (iv) *if a non-spanning component contains t rotation classes, then*

$$(8) \quad \text{pValue}(C) := 1 + \mu_C + \text{wt}(C) \geq \frac{A_k}{D_k} t.$$

Moreover, their pointwise component theorem gives the corresponding lower bound on $B_1(X)$ for every $X \in \mathcal{X}_k$.

For every component write the signed rate deviation

$$\delta(C) = D_k \text{pValue}(C) - A_k t(C) \in \mathbb{Z}.$$

For a non-spanning component, (8) implies $\delta(C) \geq 0$, and the *rate defect* is defined by

$$(9) \quad \boxed{d(C) = D_k \text{pValue}(C) - A_k t(C) \geq 0.}$$

No nonnegative defect is assigned to a spanning component. Under the standard empty-exterior convention $\mu(C) = 0$, the signed deviation of a spanning component can be negative; for example, the classical 153-letter order-five path has $\delta(C) = -17$.

A component is always a vertex-simple path. A *self-nested* preimage chain is one whose source and target are the same component; it is *self-closing* if the forced target vertex equals the component tail. A *terminal portal* will mean a component with minimum entry two and a full final interval piece. Primitive ladders and bridge blocks are defined in [Section 11](#).

3. A σ^2 -REDUCED NORMAL FORM

The two weight-two successors in (6) play different roles in the portal analysis. The required normal form is stated below.

Lemma 3.1 (Classification of weight-two successors). *Let $k \geq 3$ and $u, v \in \mathfrak{S}_k$. Then $w(u, v) = 2$ if and only if*

$$v = \sigma^2 u \quad \text{or} \quad v = \tau u.$$

Proof. Write $u = abR$. An overlap of length $k - 2$ forces v to begin with R . Its final two letters must be the two missing symbols a, b , in one of the two orders. These give $Rab = \sigma^2 u$ and $Rba = \tau u$. Both have weight exactly two because the symbols of u are distinct. $\square$

Definition 3.2. *A Hamilton path is σ^2 -reduced if it contains no edge $u \rightarrow \sigma^2 u$.*

For a path P , let $N_\sigma(P)$ be the number of its weight-one rotation edges $u \rightarrow \sigma u$.

Lemma 3.3 (σ^2 surgery). *Let $k \geq 3$. If a Hamilton path P contains $u \rightarrow \sigma^2 u$, then there is a Hamilton path P' such that*

$$\text{wt}(P') \leq \text{wt}(P), \quad N_\sigma(P') \geq N_\sigma(P) + 2.$$

Proof. Put $v = \sigma u$ and $w = \sigma^2 u$. The vertex v occurs somewhere in P . Remove it from its old position, and then insert it between the consecutive vertices u, w . Removing a vertex cannot increase path weight, by the triangle inequality for overlap weights. The insertion preserves the weight of the edge $u \rightarrow w$, because

$$w(u, w) = 2 = w(u, v) + w(v, w) = 1 + 1.$$

The resulting list is a permutation of the original Hamilton list and is therefore again a Hamilton path.

It remains to prove strict progress. At its old position, v cannot have predecessor u : the path already uses the successor w of u . Likewise, v cannot have successor w : the path already uses the predecessor u of w . Since u is the unique vertex whose rotation successor is v , and w is the unique rotation successor of v , neither old edge incident with v is a rotation edge. Deleting v therefore destroys no edge counted by N_σ , while inserting it creates the two rotation edges $u \rightarrow v$ and $v \rightarrow w$. The splice at the old location may create another rotation edge, but cannot reduce the stated gain. $\square$

Theorem 3.4 (σ^2 -reduced normal form). *For every Hamilton path P with $k \geq 3$, there exists a σ^2 -reduced Hamilton path P^* with*

$$\text{wt}(P^*) \leq \text{wt}(P).$$

Consequently, a minimum-weight Hamilton path may be chosen σ^2 -reduced.

Proof. Repeatedly apply [Lemma 3.3](#) while a σ^2 edge remains. At each step N_σ increases by at least two, while always

$$0 \leq N_\sigma(P) \leq k! - 1.$$

The process therefore terminates. Weight never increases. If P was minimum-weight, the final path has the same minimum weight. $\square$

4. THE PREIMAGE-CHAIN IDENTITY

Let P be a Hamilton path and $X = F(P)$. Hunter's construction identifies a set $S(P)$ of vertices. For every $v \in S(P)$, it removes the path edge leaving v and inserts $v \rightarrow \sigma v$; if this creates indegree two, it removes the original path edge entering the doubled vertex. Let E_1 and E_2 be the first and second sets of removed path edges, and put

$$I = \mathbf{1}[p_{k!} \in S(P)].$$

Hunter's exact bookkeeping is

$$(10) \quad \text{wt}(P) = \text{wt}(X) + \sum_{e \in E_1} (w(e) - 1) - I + \sum_{e \in E_2} w(e).$$

Define a directed graph D_P on $S(P)$ by

$$(11) \quad u \longrightarrow_{D_P} v \iff (u, \sigma v) \text{ is an edge of } P.$$

Every vertex of D_P has indegree and outdegree at most one, so D_P is a disjoint union of directed paths and cycles. Write $C_{-1}(P)$ for the rotation predecessors of the first-visited vertices of the rotation classes. The defining exclusion in Hunter's construction is

$$S(P) \cap C_{-1}(P) = \emptyset.$$

Let $\mathcal{K}(P)$ be the set of directed path components of D_P . For $K = (v_1, \dots, v_r) \in \mathcal{K}(P)$, call K *terminal* when $v_r = p_{k!}$. When K is nonterminal, its source and target Hunter components are denoted by $B(K)$ and $A(K)$, respectively.

Theorem 4.1 (Preimage-chain component matching). *Let P be a Hamilton path, $X = F(P)$, and let C_0 be the component of X containing p_1 .*

- (i) *Every $K = (v_1, \dots, v_r) \in \mathcal{K}(P)$ has a unique initial edge*

$$h_{B(K)} \longrightarrow \sigma v_1 \in E_2,$$

where $h_{B(K)}$ is the head of a unique component $B(K)$ of X .

- (ii) *Every nonterminal K has a unique exit edge*

$$v_r \longrightarrow t_{A(K)} \in E_1,$$

where $t_{A(K)}$ is the tail of a unique non-root component $A(K)$ of X .

- (iii) *If $I = 0$, there is no terminal chain. Let L be the component containing $p_{k!}$. Then*

$$B : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X) \setminus \{L\}, \quad A : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X) \setminus \{C_0\}$$

are bijections.

(iv) If $I = 1$, there is exactly one terminal chain K_* . The source map is a bijection

$$B : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X),$$

and, after setting $\hat{A}(K_*) = \star$ and $\hat{A}(K) = A(K)$ otherwise,

$$\hat{A} : \mathcal{K}(P) \xrightarrow{\sim} (\text{Comp}(X) \setminus \{C_0\}) \cup \{\star\}$$

is a bijection.

In particular, every Hunter component is the source of at most one preimage chain, and every non-root component is the target of at most one nonterminal preimage chain. This remains true for self-nested chains, for which $A(K) = B(K)$.

Proof. Let $K = (v_1, \dots, v_r)$ be a path component of D_P . Since $v_1 \in S(P)$ and $S(P)$ excludes $C_{-1}(P)$, the vertex σv_1 cannot be the first vertex p_1 . It therefore has a unique predecessor b in P . The absence of a D_P -predecessor of v_1 implies $b \notin S(P)$. Hence the original edge $b \rightarrow \sigma v_1$ survives the first deletion step, while the added edge $v_1 \rightarrow \sigma v_1$ gives σv_1 indegree two. The former edge is therefore removed in E_2 . Since $b \notin S(P)$, no added edge leaves b , and b is the head of a unique component of X . This proves the existence and uniqueness in (i).

Conversely, let h be a head of X . If $h \neq p_{k!}$, then h has a unique successor in P . It cannot lie in $S(P)$, because an added rotation edge would then leave it. Its original path edge is not removed in E_1 and must therefore be removed in E_2 . Thus it has the form $h \rightarrow \sigma v$ for a unique $v \in S(P)$. If v had a D_P -predecessor $u \in S(P)$, then the uniqueness of the predecessor of σv in P would force $u = h$, contradicting $h \notin S(P)$. Hence v is a D_P -path start. The path endpoint $p_{k!}$ is a head of X exactly when $p_{k!} \notin S(P)$, that is, exactly when $I = 0$. This proves the source bijections in (iii) and (iv).

Suppose that K is nonterminal, and let a be the successor of v_r in P . Since $v_r \in S(P)$, the edge $v_r \rightarrow a$ belongs to E_1 . If $a = \sigma u$ for some $u \in S(P)$, then $v_r \rightarrow_{D_P} u$, contrary to the maximality of K . Thus no added rotation edge enters a , and after the unique original incoming edge is removed, a is a tail of X . It is not p_1 , so it is the tail of a non-root component. This proves (ii).

Conversely, let $a \neq p_1$ be a tail of X , and let v be its unique predecessor in P . The edge $v \rightarrow a$ cannot have been removed only in E_2 , because an E_2 removal occurs precisely when an added rotation edge still enters the target, whereas a has indegree zero in X . Hence $v \rightarrow a \in E_1$, so $v \in S(P)$. Since no added edge enters a , there is no $u \in S(P)$ with $a = \sigma u$; consequently v is the endpoint of a unique D_P path. The first vertex p_1 is always an X -tail: it has no original predecessor, and an added edge $v \rightarrow p_1$ would imply $v \in C_{-1}(P) \cap S(P)$. Thus the nonterminal target map is a bijection onto $\text{Comp}(X) \setminus \{C_0\}$.

Finally, $p_{k!}$ is a vertex of D_P exactly when $I = 1$. In that case it has no D_P -successor and lies in a unique terminal path component; when $I = 0$ no terminal path exists. Directed cycles of D_P have neither starts nor ends and correspond only to internal E_1 edges, so they do not affect either matching. The stated bijections follow. $\square$

If $c = |\text{Comp}(X)|$, the theorem gives exactly $c - 1$ nonterminal chains and $c - 1 + I$ chains in total. Directed cycles have no component-level source or target and contribute only their internal E_1 edges.

For a nonterminal chain define

$$(12) \quad g(K) = [w(h_{B(K)}, \sigma v_1) - 2] + \sum_{i=1}^{r-1} [w(v_i, \sigma v_{i+1}) - 1] \\ + [w(v_r, t_{A(K)}) - \mu_{A(K)}].$$

For a terminal chain omit the last bracket. For a directed cycle $Z = (v_1, \dots, v_r)$, put

$$(13) \quad g(Z) = \sum_{i=1}^r [w(v_i, \sigma v_{i+1}) - 1], \quad v_{r+1} = v_1.$$

Every bracket is nonnegative.

Theorem 4.2 (Exact preimage-chain identity). *Let C_0 be the component of X containing the first vertex of P , and let $\mu_{\max} = \max_C \mu_C$. Then*

$$(14) \quad \boxed{\text{wt}(P) - B_1(X) = \mu_{\max} - \mu_{C_0} + I + \sum_K g(K) + \sum_Z g(Z).}$$

Proof. By Theorem 4.1, the initial edges of the D_P path components are exactly the E_2 boundary edges, there are $c - 1 + I$ of them, and the nonterminal exits are in bijection with the components $A \neq C_0$. Partition E_1 into internal chain edges, those nonterminal exits, and cycle edges. Extract a baseline two from every E_2 edge and a baseline $\mu_A - 1$ from the unique exit into every $A \neq C_0$. Including the $-I$ in (10), the extracted baseline is

$$2(c - 1 + I) + \sum_{A \neq C_0} (\mu_A - 1) - I.$$

It equals

$$(c - 1) + \sum_A \mu_A - \mu_{\max} + (\mu_{\max} - \mu_{C_0}) + I.$$

The first three terms, together with $\text{wt}(X)$, form $B_1(X)$; the remaining edge terms are precisely the quantities in (12) and (13). Directed cycles contribute only the terms $g(Z)$ and no component boundary term. $\square$

4.1. A shortest-path and assignment relaxation. Let

$$\Omega_X = \{v : (v, \sigma v) \in X\}$$

be the set of rotation slots. For a component head B , slots $u \neq v$, and a component tail A , define

$$a_B(v) = w(h_B, \sigma v) - 2, \quad b(u, v) = w(u, \sigma v) - 1, \\ c_A(v) = w(v, t_A) - \mu_A, \quad v \notin A.$$

All are nonnegative integers. Work in the extended nonnegative integers

$$\bar{\mathbb{N}} = \mathbb{N} \cup \{+\infty\}.$$

Let $d_X(B, A)$ be the minimum cost

$$a_B(v_1) + \sum_{i < r} b(v_i, v_{i+1}) + c_A(v_r)$$

over nonempty slot sequences with $v_r \notin A$; if no sequence exists, set $d_X(B, A) = +\infty$. Define $d_X(B, \star)$ similarly, omitting the exit term.

For $I = 0$, choose a candidate root R , a candidate last component L , and a bijection

$$\phi : \text{Comp}(X) \setminus \{L\} \longrightarrow \text{Comp}(X) \setminus \{R\}.$$

Set

$$\Pi_0(X) = \min \left(\mu_{\max} - \mu_R + \sum_{B \neq L} d_X(B, \phi(B)) \right).$$

For $I = 1$, choose R and a bijection

$$\phi : \text{Comp}(X) \longrightarrow (\text{Comp}(X) \setminus \{R\}) \cup \{\star\},$$

and set

$$\Pi_1(X) = 1 + \min \left(\mu_{\max} - \mu_R + \sum_B d_X(B, \phi(B)) \right).$$

Finally, $\Pi(X) = \min(\Pi_0(X), \Pi_1(X))$.

Corollary 4.3 (Preimage-chain strengthening). *For every Hamilton path P ,*

$$\text{wt}(P) \geq B_1(F(P)) + \Pi(F(P)).$$

Thus $B_1 + \Pi$ pointwise dominates B_1 and is strictly larger on explicit images, though equality may hold on record constructions.

Proof. When $I = 0$, [Theorem 4.1](#) gives bijections from the actual chains to $\text{Comp}(X) \setminus \{L\}$ at the source end and to $\text{Comp}(X) \setminus \{C_0\}$ at the target end; composing them gives a feasible assignment in the definition of Π_0 . When $I = 1$, the same theorem gives the feasible assignment to $(\text{Comp}(X) \setminus \{C_0\}) \cup \{\star\}$ required by Π_1 . The actual slot sequences are feasible routes for the corresponding d_X terms. Replacing each route by an individual shortest route, allowing different routes to reuse slots, and deleting nonnegative cycle costs can only lower the right-hand side of (14). $\square$

5. THE SIX-SYMBOL CASE: SHARPNESS AND PREIMAGE CORRECTION

The accompanying certificate consists of eight disjoint strongly exitless paths spanning $\mathfrak{S}_6$, with component sizes

$$4 \times 114, \quad 2 \times 108, \quad 2 \times 24.$$

Their weights and minimum entry weights are

multiplicity	$ C $	$\text{wt}(C)$	μ_C
4	114	134	2
2	108	127	2
2	24	26	2.

Hence

$$B_1(X) = 7 + 842 + (16 - 2) = 863.$$

The certificate verifier checks spanning, disjointness, simplicity, and strong exitlessness, so $X \in \mathcal{Y}_6$. By the formalized sufficiency direction in [Theorem 2.1](#), $X \in \mathcal{X}_6$ and therefore has at least one Hamilton preimage. The Hunter pointwise component theorem gives $B_1(Y) \geq 863$ for every $Y \in \mathcal{X}_6$. Consequently

$$(15) \quad \boxed{\min_{Y \in \mathcal{X}_6} B_1(Y) = 863.}$$

For this fixed image, the assignment relaxation gives

$$\Pi_0(X) = 6, \quad \Pi_1(X) = 7.$$

Thus every Hamilton path P with $F(P) = X$ has weight at least 869, and every word obtained from such a preimage has length at least 875. This is a statement about the fixed image X ; it is *not* the global assertion $S(6) \geq 875$.

By contrast, a known 872-letter word has path weight 866, component-size distribution

$$25 \times 24 + 1 \times 120,$$

and

$$B_1(F(P)) = 866, \quad \Pi(F(P)) = 0.$$

The correction therefore removes an independently priced false extremum while remaining exact on the known record construction.

6. RATE DEFECTS AND INTERVAL-PIECE GEOMETRY

Throughout the remainder of the structural argument, assume $k \geq 5$. The interval-piece formulas below apply to every strongly exitless component. Whenever the nonnegative rate defect $d(C)$ is used, the component is explicitly assumed non-spanning. A strongly exitless component visits each rotation class consecutively and completely. Between successive classes, a weight-two edge inside a reduced component is a proper door; maximal runs connected by such doors are called *interval pieces*. Let

- $t = t(C)$ be the number of rotation classes;
- $r = r(C)$ be the number of interval pieces;
- $\Delta = (k - 1)r - t$ be the total piece deficit;
- x be the sum over piece seams of (seam weight $- 3$);
- $\mu = \text{minto}(C)$.

A piece is *full* if it contains $k - 1$ classes; otherwise it is partial. Every class contributes $k - 1$ rotation edges, the $t - r$ within-piece doors have weight two, and the $r - 1$ seams have total weight $3(r - 1) + x$. Therefore

$$(16) \quad \text{wt}(C) = (k + 1)t + r + x - 3, \quad \text{pValue}(C) = (k + 1)t + r + x + \mu - 2.$$

Proposition 6.1 (Rate-defect normal form). *Let C be a non-spanning strongly exitless component and put $d = d(C)$. Then*

$$(17) \quad \boxed{d = D_k(x + \mu - 2) + (k - 2)\Delta - r.}$$

If

$$(18) \quad m = \Delta + (k - 1)(x + \mu - 2),$$

then

$$(19) \quad \boxed{t = D_k m - (k - 1)d, \quad \text{pValue}(C) = A_k m - k^2 d.}$$

In particular,

$$t \equiv -(k - 1)d \pmod{D_k}.$$

Proof. Substitute (16) into $d = D_k \text{pValue} - A_k t$ and use

$$A_k = (k + 1)D_k + k - 2, \quad D_k = (k - 1)(k - 2) - 1, \quad t = (k - 1)r - \Delta.$$

Solving the resulting linear equations gives (19). $\square$

7. PORTAL-RUN CAPACITY FOR INTERVAL-PIECE CHAINS

An exact weight-three chain is a sequence of interval pieces with pairwise disjoint supports and weight-three seams. A partial piece has deficit $\delta \geq 1$.

Lemma 7.1 (Portal-run local geometry). *In an exact weight-three chain:*

- (i) a pure full run has length at most $k - 2$;
- (ii) a deficit-one partial piece cannot be adjacent to positive full runs on both sides;
- (iii) every endpoint full run adjacent to a partial piece has length at most $k - 3$;
- (iv) an internal full run has length at most $k - 4$, except that length $k - 3$ is possible only when the adjacent deficits a, b satisfy $a + b \geq k - 1$.

The complete arbitrary- k coordinate proof, including all six exact weight-three successor orientations and the cyclic omitted-arc lemma, is given in [Appendix A](#). The finite program serves only as an independent transcription audit of this arbitrary- k proof.

Let a chain contain n_1 deficit-one partial pieces and n_2 partial pieces of deficit at least two. Let $e \in \{0, 1, 2\}$ be the number of chain endpoints occupied by full pieces, R the number of positive full runs, F the number of full pieces, and N the number of internal runs of length $k - 3$.

Theorem 7.2 (Endpoint-matching capacity). *For every exact weight-three chain with r pieces and total deficit Δ ,*

$$(20) \quad 2r \leq (k - 2)(\Delta + e).$$

Proof. The one-sidedness of deficit-one portals gives

$$2R - e \leq n_1 + 2n_2 \leq \Delta,$$

so $R \leq (\Delta + e)/2$. Every full run has baseline length at most $k - 4$; endpoint runs and the N long internal runs may contribute one extra full piece. Hence

$$F \leq (k - 4)R + e + N.$$

Write $p = n_1 + n_2$ and $E = \Delta - p = \sum(\delta_i - 1)$. Each long internal run consumes at least $k - 3$ units of adjacent excess, and each unit is counted by at most two neighboring runs, so $(k - 3)N \leq 2E$. Thus

$$r = p + F \leq \Delta - E + \frac{k - 4}{2}(\Delta + e) + e + \frac{2E}{k - 3}.$$

Subtracting from $(k - 2)(\Delta + e)/2$ leaves $E(k - 5)/(k - 3) \geq 0$. The pure full case has $\Delta = 0$, $e = 2$, and $r \leq k - 2$ by Lemma 7.1. $\square$

If a component has seam residual x , cut the positive-residual seams. If $a, b \in \{0, 1\}$ indicate whether its first and last pieces are full, the total number of full endpoints over the resulting chains is at most $2x + a + b$. Summing (20) yields

$$(21) \quad 2r \leq (k - 2)(\Delta + 2x + a + b).$$

8. PRIMITIVE-BLOCK STABILITY AND RIGID TERMINAL PORTALS

Definition 8.1 (Terminal primitive-block certificate). *For a component C , define*

$$(22) \quad \pi(C) = \begin{cases} 1, & \mu(C) = 2 \text{ and the last interval piece of } C \text{ is full,} \\ 0, & \text{otherwise.} \end{cases}$$

When $\pi(C) = 1$, the final full piece certifies one affine capacity unit of size D_k . This certificate is termed a terminal primitive block; it is attached to the component and does not represent an additional weak component produced by cutting the path.

Theorem 8.2 (Full primitive-block stability). *For every non-spanning strongly exitless component C ,*

$$(23) \quad 2d(C) \geq (k - 2)(m(C) - \pi(C)).$$

Equivalently,

$$(24) \quad (k - 2)t(C) \leq (k - 2)D_k\pi(C) + (D_k - 1)d(C).$$

Proof. From (17), (18), and (21),

$$\begin{aligned} 2d - (k - 2)m &= (D_k - 1)(x + \mu - 2) + (k - 2)\Delta - 2r \\ &\geq (k - 1)(k - 4)x + (D_k - 1)(\mu - 2) - (k - 2)(a + b). \end{aligned}$$

If $\mu = 2$, the first piece is partial: if it were full, both weight-two predecessors of its tail would lie in that first full structure, contradicting $\mu = 2$. Hence $a = 0$ and $b = \pi$, so the right-hand side is

at least $-(k-2)\pi$. If $\mu \geq 3$, then $\pi = 0$ and $a + b \leq 2$; the worst case $x = 0$, $\mu = 3$, $a + b = 2$ leaves

$$(D_k - 1) - 2(k - 2) = (k - 1)(k - 4) \geq 0.$$

This proves (23); substituting $t = D_k m - (k - 1)d$ gives (24). $\square$

8.1. Reduced zero-cost chains. Let P be σ^2 -reduced and let C be a component of $F(P)$ with head h_C . A zero-cost preimage chain has one slot, because every internal transition between distinct slots costs at least one.

Lemma 8.3 (Reduced zero-cost dichotomy). *Suppose a zero-cost single-slot chain leaves C and enters a component A with $\mu_A = 2$. Then its slot and target tail are forced to be*

$$(25) \quad q = \sigma^{-1}\tau(h_C), \quad t_A = \Psi(h_C) := \tau(\sigma^{-1}\tau(h_C)).$$

If the rotation class of $\Psi(h_C)$ lies in C , then either $A = C$ and $t_C = \Psi(h_C)$, or no such zero-cost chain exists.

Proof. Zero entry cost gives $w(h_C, \sigma q) = 2$. By Lemma 3.1, σq is either $\sigma^2 h_C$ or τh_C . The first alternative is an edge of the reduced path P and is excluded; hence $q = \sigma^{-1}\tau h_C$. Zero exit cost gives $w(q, t_A) = 2$. The same classification and reducedness exclude $t_A = \sigma^2 q$, leaving $t_A = \tau q = \Psi(h_C)$.

If $A \neq C$, then A contains the complete rotation class of its tail, which is already contained in C ; this contradicts disjointness of the components. If $A = C$, equality of rotation classes is insufficient: the actual target vertex must equal the component tail, namely $t_C = \Psi(h_C)$. $\square$

Lemma 8.4 (Terminal-full rigidity). *A simple component whose last piece is full cannot satisfy $t_C = \Psi(h_C)$.*

Proof. Write the start of the last full piece as Cm , where m is its marker and $|C| = k - 1$. Its head is $m\sigma^{-1}(C)$. Direct substitution in (25) gives

$$\Psi(h_C) = \sigma(C)m.$$

The last full piece contains the rotation class $[m\sigma(C)]$ and visits every vertex of that class, including $\sigma(C)m$. If this vertex were also the tail of C , the path would repeat a vertex. $\square$

Corollary 8.5 (Rigid terminal portal). *In a σ^2 -reduced Hamilton path, a component counted by $\pi(C)$ cannot enter any minimum-entry-weight-two component at zero preimage-chain cost, including itself. Its only zero-cost exceptions are being the final component or entering a component with minimum entry at least three.*

9. LAYERWISE LOWER BOUNDS

Let

$$(26) \quad B_H(k) = \mathcal{H}_k - k$$

be the Hunter path lower bound, and define

$$(27) \quad \kappa_k = D_k \left\lceil \frac{(k-2)! - (k-2)}{D_k} \right\rceil - ((k-2)! - (k-2)), \quad 0 \leq \kappa_k < D_k.$$

Lemma 9.1 (Pointwise actual-image baseline). *For every $k \geq 5$ and every Hamilton path P ,*

$$(28) \quad B_1(F(P)) \geq B_H(k).$$

Consequently

$$(29) \quad s(P) := B_1(F(P)) - B_H(k) \in \mathbb{N}, \quad B_1(F(P)) = B_H(k) + s(P).$$

Proof. The concrete image $F(P)$ belongs to the actual image class $\mathcal{X}_k$, so the pointwise Hunter–Raudvere reduction in [Theorem 2.1](#) applies to this particular image. Using the pivot strongly exitless path of length $k(k-1)(k-2)$ gives

$$B_1(F(P)) \geq k! + (k-1)! + (k-2)! - 3 + \frac{(k-2)! - (k-2)}{D_k}.$$

The left-hand side is an integer. Taking the ceiling of the final quotient yields (28), because the resulting right-hand side is exactly $\mathcal{H}_k - k$. Equation (29) follows by subtraction in $\mathbb{N}$. $\square$

For the rest of this section fix a Hamilton path P , put $X = F(P)$ and $s = s(P)$, and set

$$(30) \quad K_s = \kappa_k + sD_k.$$

Let C_0 be the actual root component containing the first path vertex. Choose a maximum- μ component C^* as *distinguished*, choosing C_0 when it is tied for the maximum. If X has one component, set $K = 0$. Otherwise every non-distinguished component is non-spanning, and K denotes the sum of their nonnegative rate defects. Write the parameters of C^* as t_*, r_*, Δ_*, x_* . Substitution in the layer equation gives

$$(31) \quad K_s - K = D_k(r_* + x_*) - (k-2)(t_* - 1).$$

Lemma 9.2 (Defect-budget monotonicity). *For every layer decomposition,*

$$(32) \quad 0 \leq K \leq K_s.$$

More precisely,

$$(33) \quad K_s - K \geq (k-1)(k-3)x_* + \frac{k-2}{2}\Delta_*.$$

Proof. Using $t_* = (k-1)r_* - \Delta_*$, rewrite (31) as

$$K_s - K = D_k x_* + (k-2)\Delta_* - r_* + (k-2).$$

By (21) and $a + b \leq 2$,

$$r_* \leq \frac{k-2}{2}\Delta_* + (k-2)x_* + (k-2).$$

Substitution gives (33). Since K is a sum of nonnegative defects, (32) follows. $\square$

Theorem 9.3 (Distinguished-component envelope). *For every layer $s \geq 0$,*

$$(34) \quad t_* \leq (k-1)(k-2) + (k-1)(K_s - K),$$

and

$$(35) \quad (k-2)t_* + (D_k - 1)K \leq C_s,$$

where

$$(36) \quad C_s = (k-1)(k-2)^2 + (D_k + 1)K_s.$$

Proof. The component-capacity bound gives

$$t_* = (k-1)r_* - \Delta_* \leq (k-1)(k-2) + (k-1)(k-2)x_* + \left(\frac{(k-1)(k-2)}{2} - 1\right)\Delta_*.$$

On the other hand, (33) implies

$$(k-1)(K_s - K) \geq (k-1)^2(k-3)x_* + \frac{(k-1)(k-2)}{2}\Delta_*,$$

which dominates the nonconstant terms above. This proves (34). Multiply by $k-2$ and add $(D_k - 1)K$. Since $(k-1)(k-2) = D_k + 1$, the coefficient of K becomes -2 , and (35) follows. $\square$

Let z be the number of non-distinguished terminal portals. Sum (24) over the non-distinguished components and combine the result with (35). This gives

$$(37) \quad (k-2)(k-1)! \leq C_s + (k-2)D_k z.$$

Define

$$(38) \quad Z_s = \max \left\{ 0, \left\lceil \frac{(k-2)(k-1)! - C_s}{(k-2)D_k} \right\rceil \right\}.$$

Then $z \geq Z_s$.

Put

$$(39) \quad E_k = (k-1)(k-3) = D_k - (k-2).$$

Lemma 9.4 (High-entry defect). *Every non-spanning component with $\mu \geq 3$ satisfies*

$$(40) \quad d \geq E_k.$$

Consequently, in layer s , the number of high-minimum-entry non-distinguished components is at most

$$\lfloor K_s / E_k \rfloor.$$

Proof. Apply (21) with $a + b \leq 2$ in (17). One obtains

$$d \geq E_k x + D_k(\mu - 2) + \frac{k-2}{2}\Delta - (k-2).$$

For $\mu \geq 3$ the right-hand side is at least $D_k - (k-2) = E_k$. The count follows from $K \leq K_s$ in Lemma 9.2. $\square$

The source correspondence incurs at most one global portal loss. If no terminal chain is present, the nonterminal source map may omit one final component. If a terminal chain is present, every component is represented by a source, but exactly one source terminates at the terminal symbol. These mutually exclusive cases therefore contribute the same single subtraction of one.

The distinguished component may provide one additional zero-cost target only when it differs from the actual root C_0 . Since ties in the maximum- μ selection are resolved in favor of C_0 , this case implies

$$\mu_{\max} - \mu_{C_0} \geq 1.$$

The exact root term in (14) pays for this additional absorption. Define

$$(41) \quad G_s = \max \left\{ 0, Z_s - 1 - \left\lfloor \frac{K_s}{E_k} \right\rfloor \right\}.$$

Theorem 9.5 (Layerwise correction for reduced paths). *Let $k \geq 5$, let P be a σ^2 -reduced Hamilton path, and let $s = s(P)$ be the canonical layer from Lemma 9.1. Then*

$$(42) \quad \boxed{\text{wt}(P) \geq B_H(k) + s(P) + G_{s(P)}}.$$

Proof. Let z be the number of non-distinguished terminal portals. By the source correspondence in Theorem 4.1, the selected nonterminal sources have codimension at most one in the component set. When $I = 0$, the omitted object may be the unique final component L ; when $I = 1$, every component is a source and the unique terminal source is the sole source without a nonterminal target. The mutual exclusivity of these cases yields one global loss of at most one portal.

If one of those nonterminal chains has zero cost, it has one slot. By Corollary 8.5, its target cannot have minimum entry two. The target map in Theorem 4.1 is injective, including in the self-nested case, so each high-minimum-entry non-distinguished component absorbs at most one zero-cost portal chain. By Lemmas 9.2 and 9.4, there are at most $\lfloor K_s / E_k \rfloor$ such targets.

The distinguished component can be one additional target only when it is not the actual root C_0 , since C_0 is excluded from the target bijection. By the tie convention, this case has $\mu_{\max} - \mu_{C_0} \geq 1$, and the root term in (14) compensates that single additional absorption. Consequently at least

$$z - 1 - \lfloor K_s / E_k \rfloor$$

portal obligations contribute a positive integer either through a chain term or through the compensating root term. Since $z \geq Z_s$, the total correction is at least G_s . Substituting the canonical equality (29) proves (42). $\square$

Corollary 9.6 (Global lower bound). *Define*

$$(43) \quad \Gamma_k = \min_{s \geq 0} (s + G_s).$$

Then, for every $k \geq 5$,

$$(44) \quad \boxed{S(k) \geq \mathcal{H}_k + \Gamma_k.}$$

Moreover,

$$(45) \quad \boxed{\Gamma_k = \left(1 - \frac{3}{k} + O(k^{-2})\right) (k-4)!}.$$

Proof. Choose a minimum-weight Hamilton path. By Theorem 3.4, it has a σ^2 -reduced minimum-weight representative P^* . Apply Theorem 9.5 to the canonical integer layer $s(P^*)$ supplied by Lemma 9.1, and then minimize the objective over all $s \geq 0$. The exact minimizer formula and asymptotic expansion are proved in Appendix C. $\square$

10. NUMERICAL LOWER BOUNDS

All quantities in Equations (27), (36), (38), (41) and (43) are integers. The values in Table 1 are checked in two complementary ways: the Lean theorem `PreimageChain.superperm_numerical_bounds_closed` derives the $S(k)$ inequalities from the unconditional global theorem, while the deterministic Python verifier reconstructs the local transition systems and independently minimizes the exact layer objective.

TABLE 1. Hunter’s certified bound, the revised correction, and the resulting computer-assisted lower bound. Upper bounds are the record 5906 at $k = 7$ and the gain-one construction for $k \geq 8$.

k	$\mathcal{H}_k$	Γ_k	new lower bound	known upper bound
5	153	0	153	153
6	869	0	869	872
7	5,888	4	5,892	5,906
8	46,103	15	46,118	46,204
9	408,338	80	408,418	408,965
10	4,032,575	505	4,033,080	4,037,046
11	43,912,566	3,669	43,916,235	43,948,807
12	522,580,501	30,263	522,610,764	522,910,088
13	6,746,243,919	279,300	6,746,523,219	6,749,568,009
14	93,887,403,944	2,852,497	93,890,256,441	93,924,230,410

For $k = 14$,

$$D_{14} = 155, \quad E_{14} = 143, \quad \kappa_{14} = 27.$$

The minimizing layer is $s = 2,852,497$. At that layer

$$K_s = 442,137,062, \quad Z_s = 3,091,864, \quad \lfloor K_s/E_{14} \rfloor = 3,091,867,$$

so $G_s = 0$. The preceding layer has objective 2,852,506, confirming strict minimality for this value of k .

11. PRIMITIVE LADDERS AND LIMITATIONS OF THE METHOD

11.1. Primitive ladders. For $0 \leq d \leq k-4$, define the *primitive ladder* $T_{k,d}$ to be the component with piece sizes

$$(k-2), (k-1)^{k-3-d}.$$

It has minimum entry two, ends in a full piece, and satisfies

$$t = D_k - (k-1)d, \quad \text{pValue} = A_k - k^2d.$$

Proposition 11.1 (Exact-weight-three primitive antichain). *No $T_{k,d}$ can be followed through an exact weight-three seam by a vertex-disjoint $T_{k,e}$.*

Proof. Put $n = k-1$. Write the source head as $mabR$, where $|R| = n-2$. The classes of its last full piece are

$$S_i = [m\sigma^i(bRa)], \quad i \in \mathbb{Z}_n.$$

An exact weight-three target start is $Rpqt$, where (p, q, t) is a permutation of (m, a, b) . The first partial and first full target pieces contain

$$P_j = [t\sigma^j(Rpq)], \quad F_j = [t\sigma^j(Rqp)].$$

A direct cyclic comparison gives

(p, q, t)	classes already shared with the source
(m, a, b)	$S_0 = F_0, \quad S_{n-1} = P_0$
(m, b, a)	$S_0 = P_{n-2}, \quad S_{n-1} = P_{n-1}$
(a, m, b)	$S_0 = P_0, \quad S_{n-1} = F_0$
(a, b, m)	$S_i = P_{i-1} \quad (i \in \mathbb{Z}_n)$
(b, m, a)	$S_0 = F_{n-2}, \quad S_{n-1} = F_{n-1}$
(b, a, m)	$S_i = F_{i-1} \quad (i \in \mathbb{Z}_n).$

Every orientation therefore shares at least two rotation classes with the source. $\square$

11.2. A two-block bridge family. A *bridge block* B_k is the exact weight-three piece chain

$$(k-3), (k-1)^{k-4}.$$

It has

$$(46) \quad P_k = k^2 - 4k + 1$$

rotation classes, defect $k-1$, minimum entry two, and a full final piece.

Proposition 11.2 (Two-block bridge family). *For every $k \geq 8$, there exists a vertex-disjoint exact weight-three successor B'_k of B_k of the same type. Their union is one strongly exitless component with*

$$t = 2P_k, \quad d = 2(k-1), \quad \pi = 1.$$

Consequently, any universal component inequality of the form

$$(47) \quad t(C) \leq D_k \pi(C) + c_k d(C)$$

must satisfy

$$(48) \quad c_k \geq \frac{k^2 - 5k + 1}{2(k-1)} = \Omega(k).$$

Proof. The explicit successor map and support-disjointness calculation are given in [Appendix B](#). Concatenation adds one weight-three seam and preserves minimum entry two. The combined piece parameters are $r = 2(k - 3)$, $\Delta = 4$, $x = 0$, and $\mu = 2$, giving the displayed t, d, π . Substitution in (47) yields (48). $\square$

Thus the explicit family proves that a universal single-component linear envelope of the specific form (47) cannot have $c_k = O(1)$. It does not exclude nonlinear defect functions, multiple portal states per component, prefix-sensitive potentials, or joint inequalities coupling several components.

11.3. Contraction under neutral refinement.

Proposition 11.3 (Weight-three neutral refinement). *Suppose q internal weight-three edges are cut inside minimum-entry-weight-two components, every resulting block still has minimum entry two, and the global maximum minimum-entry value is unchanged. Then the refined graph has the same value of B_1 . Moreover, the removed edges form a zero-surplus matching saturating every refined block except one per original component.*

Proof. The total edge weight changes by $-3q$, the number of components by $+q$, and the sum of minimum-entry values by $+2q$. Hence B_1 changes by $-3q + q + 2q = 0$. Within each original component, the removed edges join the head of each block to the tail of the next. Relative to the refined baseline, their surplus is $3 - (1 + 2) = 0$, and their sources and targets are distinct. They match all but one block per original component. $\square$

The proposition excludes the derivation of a new factorial level by cutting existing weight-three seams and counting the resulting blocks independently. Any improvement of this type requires higher-order integer packing constraints or a non-neutral second transform.

12. FORMAL VERIFICATION, FINITE AUDITS, AND REPRODUCIBILITY

The immutable ancillary archive accompanying this version has SHA-256

8ce335b549b5c52c4ffbbe91f1a75528ebdbd8d9e3de49d21002c32d14a0dd65.

The archive is versioned as 4.0.1-lean-formalized-release. The formal development uses Lean 4.31.0 and pins the Hunter–Raudvere baseline to

d45222190031d162feb1f6f3cf5fe2d11fab726d.

The public formalization repository is the

[public GitHub repository](#),

and the exact formalization source snapshot is commit

a3852f0de2d84b99cc65d4c281e86892b5f6e911.

The gain-one upper-bound source remains pinned to

62958c6268f728828ed578a171486866b6fe4b20.

Lean theorem boundary. The public root module exports the following final theorems.

Lean theorem	Mathematical role
<code>actual_reduced_pathwise_bound_closed</code>	The corrected lower bound for every σ^2 -reduced Hamilton path.
<code>pathwise_new_bound_closed</code>	The unconditional pathwise theorem, obtained by the non-increasing σ^2 normal form.
<code>superperm_new_bound_closed</code>	The unconditional global lower bound after the Hamilton-path-to-superpermutation bridge.
<code>superperm_numerical_bounds_closed</code>	Direct $S(k)$ inequalities for every $k = 5, \dots, 14$.

The release head was checked with

```
lake --rehash build
lake env lean AxiomCheck.lean
rg -n '\bsorry\b|\badmit\b|\^s*axiom\b' \
  --glob '*.lean' --glob '!vendor/**' --glob '!proof-progress/**' .
```

The rehashed build completed successfully with 8714 jobs, and the active-source scan returned no match. The four exported final theorems use only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`; no project-local axiom is introduced.

The formalization covers the actual component-capacity chain, primitive-block stability, the pointwise actual-image baseline of [Lemma 9.1](#), the root-biased distinguished-component budget, the single-exception capacity-one portal assignment, the σ^2 reduction, the global theorem, and the direct numerical corollaries. The foundational Hunter–Raudvere definitions and theorems are imported from the pinned external library.

Independent deterministic finite audits. A separate deterministic Python audit reproduces the finite transcriptions and certificate checks using only the Python standard library.

File	Purpose
<code>verify_reduced_weight2_normal_form.py</code>	Audits the two weight-two successors and the local monotone-measure identities used in the σ^2 -reduced normal form.
<code>verify_hunter_bound1_sharpness_n6.py</code>	Checks the eight-component order-six certificate: spanning, disjointness, simplicity, strong exitlessness, weights, minimum entry values, and $B_1 = 863$.
<code>verify_preimage_chain_bound_n6.py</code>	Reconstructs $F(P)$, checks the exact identity on nineteen order-four paths, evaluates Π , and calibrates the two order-six examples.
<code>verify_preimage_component_matching.py</code>	Independently reconstructs $F(P)$ and D_P ; exhausts all 720 order-three Hamilton paths and checks archived plus deterministic order-four samples against the source/target component bijections.
<code>verify_full_primitive_block_stability.py</code>	Audits local portal states, primitive transitions, the exact integer layer formula, and the numerical table through order fourteen.
<code>verify_revision_edge_cases.py</code>	Checks the spanning signed-deviation example, the $k = 5$ bridge ambiguity, and the explicit $k = 8, \dots, 14$ bridge construction.
<code>verify_self_nesting_bridge_macro_theory.py</code>	Calibrates self-nesting, terminal-full rigidity, primitive rigidity, and supplementary bridge-macro experiments.

The canonical finite-audit commands are

```
python run_quick_audit.py
python run_full_audit.py
```

The frozen reference environment is Linux `x86_64` with CPython 3.13.5 and no third-party Python packages. Exact commands, claim-to-checker mapping, expected reports, release meta-data, final Lean logs, and SHA-256 manifests are included in the ancillary archive.

Layer	Status
Hunter–Raudvere baseline	Pinned external Lean 4.31.0 development
New arbitrary- k main proof chain	Lean 4.31.0 formalization in the merged release
Order-six certificates and $k \leq 14$ finite computations	Deterministic Python reconstruction and audit
Independent replication	Outside the scope of the present verification record

13. CONCLUSIONS AND OUTLOOK

The preimage-chain identity quantifies the ordering information lost when Hunter components are priced independently. The σ^2 -reduced normal form removes the ambiguity among zero-cost weight-two transitions, while the rate-defect normal form and endpoint-matching capacity theorem convert a bounded defect budget into a family of rigid terminal portals. The pointwise actual-image baseline assigns a canonical nonnegative layer to every Hamilton path, and the capacity-one portal assignment accounts for all but one global source/terminal exception. These ingredients yield the explicit correction Γ_k and the lower bounds in [Table 1](#).

The proof from actual component geometry to the unconditional global $S(k)$ theorem is verified in Lean 4.31.0. The formalization confirms that the single subtraction in G_s is shared by the two mutually exclusive terminal cases and that a distinguished non-root zero-cost target is charged to the exact root gap $\mu_{\max} - \mu_{C_0}$.

The six-symbol analysis distinguishes a low-priced component system from a low-priced Hamilton preimage: the explicit image has $B_1 = 863$, but each of its preimages incurs at least six additional units in the assignment relaxation. This observation does not imply $S(6) \geq 875$, since the known 872-letter construction satisfies $B_1 + \Pi = 866$.

The bridge construction and the neutral-refinement contraction delimit a precise subclass of the method. A single-component, single-portal affine envelope cannot have a bounded defect coefficient, and cutting existing weight-three seams alone creates no new independent obligation. The obstruction does not apply to nonlinear, multistate, intercomponent, prefix-sensitive, or iterated-transform methods. Any improvement beyond the present $\Theta((k-4)!)$ correction therefore requires additional higher-order structure.

AUTHOR CONTRIBUTIONS

Conceptualization, methodology, formal analysis, software, validation, investigation, writing—original draft, and writing—review and editing: Xiaolong Liu.

CONFLICT OF INTEREST

The author declares no conflict of interest.

DATA AND CODE AVAILABILITY

The ancillary release identified in [Section 12](#) contains the deterministic verifiers, compact certificates, canonical reports, exact commands, claim maps, the final Lean release record, selected proof sources, build and axiom-audit logs, and SHA-256 manifests. The formalization is identified by the [public repository](#) and source snapshot `a3852f0de2d84b99cc65d4c281e86892b5f6e911`. The repository also provides the English, Chinese, and bilingual papers and the ancillary archive.

DECLARATION ON GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, ChatGPT Pro (OpenAI) was used for exploratory derivations, computational prototyping, Lean and Python code development, and language editing. All mathematical claims, proofs, citations, source code, and conclusions were reviewed by Xiaolong Liu, who assumes full responsibility for the published work. The Lean formalization and deterministic computational audits are documented in [Section 12](#); they do not constitute independent peer review.

APPENDIX A. COORDINATE PROOF OF THE PORTAL LEMMAS

Put $n = k - 1$. A piece start is a word $z = z_0 \cdots z_{k-1}$; its marker is z_{k-1} and its cyclic base is $z_0 \cdots z_{k-2}$. A piece of size s contains

$$(49) \quad \mathcal{P}(z, s) = \{[z_{k-1}\sigma^i(z_0 \cdots z_{k-2})] : 0 \leq i < s\}.$$

Its endpoint is $z_{k-1}\sigma^{s-1}(z_0 \cdots z_{k-2})$. An exact weight-three successor retains the suffix after the first three endpoint letters and appends one of their six permutations.

Lemma A.1 (Marked-class alignment). *Let W, W' be permutation words on the same alphabet and let z be a designated symbol. Each rotation class has a unique representative beginning with z . Consequently, $[W] = [W']$ if and only if the representatives obtained by rotating W and W' to place z first are identical letter by letter.*

Proof. The symbol z occurs exactly once in each permutation word, so exactly one cyclic rotation begins with z . Equality of rotation classes is therefore equivalent to equality of these unique representatives. $\square$

A.1. Full runs. Write a full-piece start as Yam , with $|Y| = k - 2$. Direct substitution in (49) shows that its unique support-disjoint full successor is

$$(50) \quad Q(Yam) = \sigma(Y)am.$$

The supports of $z, Qz, \dots, Q^{k-3}z$ are pairwise disjoint. Equality of a class from $Q^i z$ and one from $Q^j z$ aligns the unique letters a, m and then forces $i = j$ modulo $k - 2$. The next support repeats the first. This proves [Lemma 7.1\(i\)](#).

A.2. All six successor orientations at a partial endpoint. Normalize a partial piece of deficit δ so that its endpoint is $mabR$, where $|R| = k - 3$. In the marked complete-two-cycle with marker m and cyclic base abR , label incidences by $\mathbb{Z}_n$ so that the endpoint class is 0. The piece occupies

$$(51) \quad I_\delta = \{0, \delta + 1, \delta + 2, \dots, n - 1\};$$

the omitted cyclic arc is $J_\delta = \{1, \dots, \delta\}$.

The six exact weight-three successor starts are $R\pi(m, a, b)$. If $\pi(m, a, b) = (p, q, t)$, then after j applications of the full-piece map Q the candidate full piece has start

$$\sigma^j(Rp)qt$$

and support

$$(52) \quad \mathcal{F}_j(p, q, t) = \{[t\sigma^\ell(\sigma^j(Rp)q)] : \ell \in \mathbb{Z}_n\}.$$

By [Lemma A.1](#), an intersection can be tested by rotating both words to put the same distinguished letter first. Since the letters of R retain their cyclic order, once the positions of m, a, b are fixed the remaining letters force a unique pair of incidence indices (i, ℓ) . Solving these six

elementary word equalities gives the following complete table. The integer j is the index of the prospective full piece, beginning with $j = 0$; all incidence indices are taken modulo n .

(p, q, t)	$j_{\min}$	one forced equality (i, ℓ)	disjoint full pieces
(m, a, b)	0	$(0, 0)$	0
(m, b, a)	0	$(0, n - 1)$	0
(a, b, m)	0	$(\delta + 1, \delta - 1)$	0
(a, m, b)	$\delta - 1$	$(\delta + 1, n - \delta)$	$\delta - 1$
(b, m, a)	$\delta - 1$	$(\delta + 1, n - \delta - 1)$	$\delta - 1$
(b, a, m)	$n - 2$	$(\delta + 1, \delta)$	$n - 2 = k - 3$.

The notation (i, ℓ) denotes

$$[m\sigma^i(abR)] = [t\sigma^\ell(\sigma^{j_{\min}}(Rp)q)].$$

For the first three orientations the displayed equality already occurs at $j = 0$. For (a, m, b) and (b, m, a) , alignment of the ordered subword R forces $j \geq \delta - 1$, and the displayed pairs show equality at $j = \delta - 1$; thus no earlier full piece intersects. For (b, a, m) , the same alignment forces $j \equiv n - 2 \pmod{n}$, so the candidates $j = 0, \dots, n - 3$ are disjoint and $j = n - 2$ is the first return. These necessity statements exclude every other rotation equivalence, while the displayed equalities prove sufficiency. This proves the endpoint bound in Lemma 7.1(iii).

When $\delta = 1$, the fourth and fifth rows have length zero. The only positive full run uses the last orientation and its boundary full piece contains the unique omitted incidence. A positive full run on the other side of the same unit partial would have to use that same incidence. The two boundary full pieces would overlap, proving Lemma 7.1(ii).

A.3. The internal long-run threshold. Let the left partial deficit be a . Follow the unique long orientation for $n - 2 = k - 3$ full pieces. Normalize the resulting head as $macU$. The six possible right-partial starts are $U\pi(m, a, c)$. Five orientations have an unavoidable common class:

$\pi(m, a, c)$	explicit class already in the accumulated support
(m, a, c)	$[macU]$
(m, c, a)	$[macU]$
(a, m, c)	$[mcUa]$
(a, c, m)	$[mUac]$
(c, m, a)	$[maUc]$.

The remaining orientation is (c, a, m) . Its marker is m . After transporting both boundary pieces to the same marked incidence cycle, choose coordinates so that the left occupied interval is

$$L_a = \{0, a + 1, a + 2, \dots, n - 1\}$$

and the right occupied interval is

$$R_b = \{1, 2, \dots, n - b\}.$$

Thus

$$L_a \cap R_b = \{a + 1, a + 2, \dots, n - b\}, \quad |L_a \cap R_b| = \max(0, n - a - b).$$

The two boundary pieces are disjoint exactly when

$$(53) \quad a + b \geq n = k - 1.$$

The preceding five orientations already have the explicit common classes listed above, so (c, a, m) is the unique orientation that can be disjoint. This proves Lemma 7.1(iv), including both necessity and sufficiency of the threshold.

APPENDIX B. AN EXPLICIT TWO-BLOCK BRIDGE CONSTRUCTION

Assume $k \geq 8$ and put $n = k - 1$. Let a labeled base be $B = (b_0, \dots, b_{n-1})$, with marker m , and set $a = b_{n-3}$. Start with the partial piece at Bm of size $k - 3 = n - 2$. No uniqueness is required for the transition from this partial piece to the first full piece. An explicit choice is made with start

$$(54) \quad z_1 = (b_{n-1}, b_0, \dots, b_{n-4}, b_{n-2}, b_{n-3}, m).$$

After this choice, take the remaining $k - 5$ full pieces successively by the unique full-to-full map Q from (50). The resulting chain, of piece sizes $(k - 3), (k - 1)^{k-4}$, is the bridge block $B_k(B, m)$.

A direct endpoint calculation gives an exact weight-three successor start with the same marker and base

$$(55) \quad \Theta(B) = (b_{n-4}, b_{n-2}, b_{n-1}, b_0, \dots, b_{n-5}, b_{n-3}),$$

where all displayed index ranges are ordinary ranges because $n \geq 7$.

A direct support argument follows. Because the marker m is unique, two classes written with m first are equal exactly when their length- n bases are equal. Delete the anchor $a = b_{n-3}$ from the source bridge cycles. Every source piece is obtained from the fixed cyclic skeleton

$$L = (b_0, b_1, \dots, b_{n-4}, b_{n-2}, b_{n-1})$$

by inserting a in one of its cyclic gaps, except for the single omitted gap

$$(56) \quad (b_{n-5}, b_{n-4}).$$

Indeed, the first partial inserts a between b_{n-4} and b_{n-2} , while the successive full pieces insert it, in order, between

$$(b_{n-2}, b_{n-1}), (b_{n-1}, b_0), (b_0, b_1), \dots, (b_{n-6}, b_{n-5}).$$

For the target base $B' = \Theta(B)$, its varying anchor is $a' = b_{n-6}$. Delete a' and rotate the remaining skeleton to begin with b_0 ; the result is

$$L' = (b_0, \dots, b_{n-7}, b_{n-5}, a, b_{n-4}, b_{n-2}, b_{n-1}).$$

After deleting both a and a' , the source and target have the same common skeleton. In every target bridge cycle the letter a is fixed in the gap (b_{n-5}, b_{n-4}) , because only a' varies among the target gaps. Therefore a target cycle could equal a source cycle only if the source inserted a in the gap (56); that is precisely the one gap absent from the source block. Hence the two bridge supports are disjoint.

The five other permutations of the three endpoint letters already share classes with the source last-full support, as in the six-orientation comparison of Proposition 11.1. Thus (55) supplies the required disjoint successor.

The two blocks together have $r = 2(k - 3)$, $\Delta = 4$, $x = 0$, and $\mu = 2$. Equations (17) and $t = (k - 1)r - \Delta$ give the parameters used in Proposition 11.2.

APPENDIX C. EXACT MINIMIZATION OF THE LAYER CORRECTION

Put

$$M = (k - 1)!, \quad D = D_k, \quad E = (k - 1)(k - 3).$$

Let Z_0 be (38) at $s = 0$. Since $D + 1 = (k - 1)(k - 2)$ and $C_s = C_0 + (D + 1)Ds$,

$$(57) \quad \boxed{Z_s = \max(0, Z_0 - (k - 1)s)}.$$

Also $D = E + (k - 2)$, so

$$(58) \quad \left\lfloor \frac{\kappa_k + sD}{E} \right\rfloor = s + \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor.$$

Consequently

$$(59) \quad G_s = \max \left(0, Z_0 - 1 - ks - \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor \right).$$

Proposition C.1 (Exact minimizing layer). *The value Γ_k is the least integer $s \geq 0$ such that*

$$(60) \quad ks + \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor \geq Z_0 - 1.$$

Proof. While $G_s > 0$, equations (43) and (59) give

$$s + G_s = Z_0 - 1 - (k - 1)s - \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor,$$

which strictly decreases with s . Once $G_s = 0$, the objective equals s and strictly increases. At the preceding layer, failure of (60) is an integer failure by at least one, so its objective is no smaller than the first zero-gap layer. $\square$

Finally,

$$(61) \quad Z_0 = \frac{M}{D} + O(k),$$

because $0 \leq \kappa_k < D$. From (60),

$$\frac{Z_0 - 1 - \kappa_k/E}{k + (k - 2)/E} \leq \Gamma_k \leq \left\lceil \frac{Z_0 - 1}{k} \right\rceil.$$

Thus

$$(62) \quad \Gamma_k = \frac{(k - 1)!}{kD_k} + O(1) = \left(1 - \frac{3}{k} + O(k^{-2}) \right) (k - 4)!,$$

since

$$\frac{(k - 1)!}{kD_k(k - 4)!} = \frac{(k - 1)(k - 2)(k - 3)}{k(k^2 - 3k + 1)} = 1 - \frac{3}{k} + O(k^{-2}).$$

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